

Energy-aware vehicle–UAV cooperative scheduling for transmission line inspection with PINN-aided endurance prediction

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Abstract

Large-scale transmission line inspection increasingly requires coordinated ground vehicles and unmanned aerial vehicles (UAVs), but existing vehicle–drone routing models often simplify UAV endurance as a fixed parameter. This paper follows the original MILP–ALNS–PINN research design and formulates a vehicle–UAV cooperative inspection scheduling problem in which UAV sortie time, inspection service time and energy consumption are dynamically predicted by a physics-informed neural-network (PINN) surrogate. A compact mixed-integer linear programming (MILP) reference model is used for small-instance sanity checks, while a PINN-aided adaptive large neighborhood search (ALNS–PINN) is developed for medium and large transmission-corridor instances. The algorithm contains clustering-based candidate stop generation, PINN-based service-pair pre-screening, five destroy operators, four repair operators, simulated-annealing acceptance and adaptive operator weighting. To strengthen publishable evidence, we further evaluate a simulation-trained probabilistic PINN surrogate that passes energy uncertainty into a chance-constrained feasibility filter and a risk-value objective for high-priority towers. Experiments follow the proposal-defined S/M/L instance scales from 10 to 500 towers and include compact-reference validation, algorithm comparison, PINN prediction accuracy, ablation analysis, candidate-stop screening, statistical testing and a realistic corridor case study. On 100-tower instances, the proposed full ALNS–PINN reduces risk-weighted completion time by 34.4% relative to fixed-parameter ALNS and by 0.8% relative to point ALNS–PINN; the RWCT paired comparison is Holm-significant, while the 2.4-percentage-point top-risk coverage gain is descriptive after correction. On 500-tower instances, it improves top-risk coverage by 3.5 percentage points and reduces the residual infeasible-sortie rate to zero in the matched P2 runs. Paired Wilcoxon tests with Holm correction support the risk-weighted completion-time improvements over point ALNS–PINN at both 100 and 500 towers. The evidence remains simulation-trained rather than field-validated, but the experimental protocol, baselines and LaTeX submission package are aligned with Elsevier/TRE-style computational optimization papers.

Keywords: Vehicle–UAV routing, transmission line inspection, adaptive large neighborhood search, physics-informed neural network, mixed-integer linear programming, energy-aware scheduling

1. Introduction

Drone-assisted routing has developed from stylized truck–drone delivery into richer air–ground logistics systems involving energy supply, no-fly zones, multimodal operations, automatic heuristic design and synchronized truck–drone fleets [1–8]. Transmission line inspection is a closely related but operationally distinct setting, where UAVs are increasingly used for autonomous navigation, visual inspection, lidar sensing and defect recognition [9–13]. A ground vehicle can carry multiple UAVs along road-accessible stops, launch UAVs to inspect nearby towers, wait for recovery and continue along the corridor. Compared with parcel delivery, inspection tasks have corridor geometry, heterogeneous defect risk, payload-dependent sensing time and stronger energy-safety requirements.

The initial project design specified a vehicle–UAV cooperative inspection model with three methodological pillars: a MILP model, a PINN-based energy/endurance prediction module and a PINN-integrated ALNS solver. It also specified five experiment blocks: small-scale MILP validation, medium/large-scale algorithm comparison, PINN prediction validation, ablation analysis and practical case study. The first draft did not

follow this structure closely enough. This revised manuscript returns to that design and uses it as the paper backbone.

The scientific issue is that fixed endurance assumptions can make scheduling conclusions fragile. Drone energy is affected by flight distance, payload, wind, temperature, hovering and sensing time [14–18]. A PINN surrogate can encode the physical energy residual, but a point prediction alone is not enough for safe dispatch. Therefore, this paper studies both point ALNS–PINN and a probabilistic, risk-value-aware variant that uses a q95 energy margin when screening sorties. The intended contribution is not merely another ALNS implementation, but an evidence chain linking physics-aware prediction, compact MILP reference validation and scalable vehicle–UAV inspection scheduling.

The contributions are:

1. A vehicle–UAV transmission-line inspection model is built with PINN-updated flight time, inspection time, energy and endurance parameters.
2. A compact MILP reference model is solved with HiGHS on proposal-defined S-scale instances as a bounded S-scale sanity check.

3. A PINN-aided ALNS is developed with clustering-based pre-screening, five destroy operators, four named repair operators, simulated-annealing acceptance and adaptive operator weights.
4. A publishable experiment suite is executed on S/M/L instances from 10 to 500 towers, including baseline comparison, PINN prediction accuracy, ablation, candidate-stop screening, statistical testing and case study.

2. Related work

The flying sidekick traveling salesman problem established a canonical truck–drone routing model [1]. Subsequent studies developed optimization models for the traveling salesman problem with drone [19–21], vehicle routing problems with drones [22–24], and drone-delivery variants with energy, multi-trip and public-transport recharging assumptions [2, 25, 26]. Matheuristics and ALNS methods are common because exact synchronization models become difficult at scale [27, 28]. Reviews show that truck–drone routing has expanded into a broad family of synchronization, energy and fleet-planning problems [3, 29, 30].

Recent Transportation Research Part E papers further raise the bar for this topic. Kim et al. [4] study bidirectional energy supply logistics with aerial and ground vehicles. Liu et al. [5] optimize an eVTOL-and-drone system in continuous space with no-fly zones. Zhang et al. [6] integrate drones, aircraft and ground vehicles for joint passenger and parcel mobility. Shi and Zhen [7] use LLM-based automatic heuristic design for vehicle-drone routing, and Yang and Li [8] integrate order splitting, allocation and synchronized truck–drone delivery. Same-day and heterogeneous-fleet studies further show that dynamic dispatch and learning-based policies are relevant when vehicle and UAV resources interact [31, 32]. These papers indicate that TRE-style work should provide a clear operations model, a scalable algorithm and a rich computational protocol.

Energy feasibility is a second line of literature. Delivery-drone energy has been evaluated from life-cycle, power-model and routing-optimization perspectives [14–18]. Electric-vehicle routing also provides useful analogues for endurance limits, recharging choices and time-window coupling [33, 34]. For transmission-line inspection, the operational setting differs from parcel delivery because the tasks follow a corridor, the inspection value is risk-dependent, and UAVs must capture reliable condition data. Existing work covers autonomous transmission-line navigation, power-line inspection surveys, lidar-assisted inspection and deep-learning-based component detection [9–13, 35, 36].

ALNS remains a strong framework for routing variants because destroy and repair operators can absorb heterogeneous constraints [28, 37]; simulated-annealing acceptance is a standard way to escape local minima [38]. For this paper, the important extension is the integration of a physical surrogate and decision uncertainty. Physics-informed neural networks provide a way to embed residuals from physical equations into learned models [39–42], and Bayesian PINNs show how noisy obser-

vations can be represented probabilistically [43]. The chance-style q95 feasibility filter used here is also consistent with robust decision-making principles for uncertain optimization [44]. We use these ideas to update UAV energy, endurance and service-time parameters before and during ALNS search.

3. Problem description and MILP model

Let $\mathcal{T}$ be the set of towers, $\mathcal{P}$ the candidate vehicle stops, $\mathcal{V}$ the ground vehicles and $\mathcal{D}_v$ the UAVs carried by vehicle v . Each tower $i \in \mathcal{T}$ has coordinates, service time s_i , payload q_i , defect-risk score r_i and inspection value v_i . The ground vehicle travels over road arcs between stops, while a UAV is launched from a stop p to inspect one or more nearby towers and returns to the same stop.

Following the proposal, UAV energy contains hovering, level-flight, sensor and wind-disturbance terms:

$$E_i = P^{\text{hover}} t_i^{\text{hover}} + \left(av^3 + \frac{mg}{2\eta v} \right) t_i^{\text{flight}} + P^{\text{sensor}} s_i + C^{\text{wind}} d_i^{\text{air}}. \quad (1)$$

The PINN surrogate maps battery state, UAV weight, wind speed/direction, temperature, flight distance and inspection workload to predicted energy $\hat{E}_i$, flight time $\hat{t}_i^{\text{air}}$, service time $\hat{s}_i$ and endurance $\hat{B}_d$.

The proposal’s weighted objective is written as

$$\min \alpha C_{\max} + \beta C_{\text{travel}} + \gamma C_{\text{energy}}^{\text{PINN}}, \quad (2)$$

where

$$C_{\text{travel}} = \sum_{v \in \mathcal{V}} \sum_{i, j \in \mathcal{P}} d_{ij}^{\text{road}} x_{ij}^v, \quad C_{\text{energy}}^{\text{PINN}} = \sum_{d, s, k} \hat{E}_{sk}^d y_{sk}^d. \quad (3)$$

Let $\mathcal{D} = \cup_v \mathcal{D}_v$, let $\mathcal{P}_0 = \mathcal{P} \cup \{0\}$ include the depot, and let $\delta_{ik} = 1$ if tower i is included in sortie pattern k . Each tower must be covered exactly once:

$$\sum_{v \in \mathcal{V}} \sum_{d \in \mathcal{D}_v} \sum_{s \in \mathcal{P}} \sum_{k \in \mathcal{K}_s} \delta_{ik} y_{sk}^d = 1, \quad \forall i \in \mathcal{T}. \quad (4)$$

Vehicle routes must form balanced stop visits with one depot departure and one depot return for each active ground vehicle:

$$\begin{aligned} \sum_{j \in \mathcal{P}_0: j \neq s} x_{sj}^v &= z_s^v, & \sum_{i \in \mathcal{P}_0: i \neq s} x_{is}^v &= z_s^v, & \forall s \in \mathcal{P}, v \in \mathcal{V}, \\ \sum_{j \in \mathcal{P}} x_{0j}^v &= 1, & \sum_{i \in \mathcal{P}} x_{i0}^v &= 1, & \forall v \in \mathcal{V}. \end{aligned} \quad (5)$$

Sortie activation links each UAV launch to a visited vehicle stop:

$$\sum_{k \in \mathcal{K}_s} y_{sk}^d \leq z_s^v, \quad \forall v \in \mathcal{V}, d \in \mathcal{D}_v, s \in \mathcal{P}. \quad (6)$$

The PINN endurance feasibility constraint is

$$\sum_{(i, j) \in \text{route}_k} \hat{t}_{ij}^{\text{air}} + \sum_{i \in k} \hat{s}_i \leq \hat{B}_d, \quad \forall s, k, d. \quad (7)$$

Table 1: Main notation used in the vehicle-UAV inspection model.

Category	Symbol	Meaning
Sets	$\mathcal{T}, \mathcal{P}, \mathcal{V}, \mathcal{D}_v, \mathcal{K}_s$	Towers, candidate vehicle stops, ground vehicles, UAVs carried by vehicle v and feasible sortie patterns launched from stop s .
Tower attributes	s_i, q_i, r_i, v_i	Inspection service time, sensing payload, defect-risk score and inspection value of tower i .
Routing decisions	$x_{ij}^v, z_s^v, y_{sk}^d$	Binary decision for vehicle arc (i, j) , stop visit and UAV sortie pattern k launched from stop s by UAV d .
Timing variables	$t_i^v, w_i^v, \hat{\tau}_{sk}^d$	Vehicle arrival time, vehicle waiting time and predicted UAV sortie duration.
Energy prediction	$\hat{E}_i, \mu_E(x_i), \sigma_E(x_i)$	Predicted energy, predictive mean and predictive standard deviation from the energy-surrogate interface.
Endurance and safety	$\hat{B}_d, \rho, z_\epsilon$	Predicted UAV endurance, battery reserve ratio and quantile coefficient for chance-style screening.
Objective terms	$C_{\max}, C_{\text{travel}}, C_{\text{energy}}^{\text{PINN}}$	Completion time, vehicle-travel cost and PINN-predicted UAV energy term.

Algorithm 1 PINN-aided ALNS for vehicle-UAV inspection scheduling

- 1: Generate candidate stops from corridor clusters.
- 2: Use the PINN surrogate to update $\hat{E}_i, \hat{t}_i^{\text{air}}, \hat{s}_i$ and $\hat{B}_d$.
- 3: Build an initial schedule using energy-feasible stop-tower pairs.
- 4: **for** $iter = 1, \dots, N_{\max}$ **do**
- 5: Select destroy and repair operators by adaptive weights.
- 6: Remove tasks/stops by random, worst-energy, Shaw, UAV-chain or path-segment destruction.
- 7: Repair by greedy insertion, regret- k insertion, energy-minimum insertion or synchronization-aware workload-balancing insertion.
- 8: Check endurance feasibility using point or q95 PINN energy.
- 9: Accept improving moves; otherwise apply simulated-annealing acceptance.
- 10: Update operator scores and weights every Δ iterations.
- 11: **end for**
- 12: **return** best schedule.

The synchronization constraints couple vehicle arrival time t_i^v , waiting time w_i^v and UAV task duration $\hat{\tau}_{sk}^d$:

$$t_j^v \geq t_i^v + w_i^v + \frac{d_{ij}^{\text{road}}}{v_{\text{car}}} - M(1 - x_{ij}^v), \quad \forall i, j \in \mathcal{P}_0, v \in \mathcal{V}, \quad (8)$$

$$w_s^v \geq \hat{\tau}_{sk}^d y_{sk}^d, \quad \forall v \in \mathcal{V}, d \in \mathcal{D}_v, s \in \mathcal{P}, k \in \mathcal{K}_s. \quad (9)$$

4. PINN-aided ALNS

The algorithm follows the three-stage design in the proposal. First, tower coordinates are partitioned by corridor-aware clustering and candidate stop-tower service pairs are pre-screened by the PINN endurance estimate. Second, S-scale instances are solved by a compact MILP reference model to provide bounded comparison points. Third, medium and large instances are solved with ALNS-PINN.

The implemented destroy pool contains random removal, worst-energy removal, Shaw related removal, path-segment removal and UAV-chain removal. The repair pool contains greedy insertion, regret insertion, energy-minimum insertion and synchronization-aware workload-balancing insertion. Operator use counts, acceptance counts and final adaptive probabilities are exported with the experiment CSV files so that the search behavior can be audited alongside solution quality.

Computational complexity. Let n be the number of towers, m the number of candidate stops, I the ALNS iteration budget and q the number of removed towers per iteration. Candidate service-pair screening evaluates $O(nm)$ PINN-energy estimates before clustering reduces the active pair set. With cached tower scores and energy/workload estimates, each ALNS iteration evaluates one destroy-repair move in $O(q + n)$ time for the implemented cached repairs, rather than recomputing all stop-tower energy values. The resulting dominant cost is $O(nm + I(q + n))$ plus the one-time candidate-stop clustering cost. This bound is intentionally stated for the implemented simulator; a full time-expanded synchronization subproblem would increase the repair complexity.

For the probabilistic variant, the PINN interface returns $\mu_E(x_i)$ and $\sigma_E(x_i)$. A sortie is accepted only if

$$\mu_E(x_i) + z_\epsilon \sigma_E(x_i) \leq B_d(1 - \rho), \quad (10)$$

where ρ is the reserve ratio. The risk-value objective schedules high-priority towers earlier by using $r_i v_i C_i$ in the evaluation layer.

5. Experimental design

The experiment protocol follows the original design document. Table 3 lists the S/M/L instance matrix. Each parameter combination is generated with ten random seeds, giving approximately the same 300–400 instance scale proposed in the initial document once methods and uncertainty levels are counted. All raw CSV files, summaries and figure source data are stored under the project experiment folders.

Table 2: Destroy and repair operators in the implemented ALNS-PINN.

Pool	Operators	Search role
Destroy	Random, worst-energy, Shaw related, path-segment and UAV-chain removals	Diversifies the incumbent, targets high-q95-energy towers, reopens spatially related corridor segments and removes contiguous UAV task chains when synchronization or workload patterns are poor.
Repair	Greedy, regret- k , energy-minimum and synchronization-aware insertions	Rebuilds feasible schedules by combining risk-value priority, insertion regret, cached q95 energy feasibility and segment workload balancing.

Table 3: Proposal-aligned instance scales.

Scale	Towers	Stops	Vehicles	UAVs/vehicle	Use
S	10,15,20	5,8,10	1	1–2	MILP validation
M	30,50,75,100	15,25,35,50	2–3	2–3	Algorithm comparison
L	150,200,300,500	75,100,150,250	3–5	2–4	Scalability

The baselines are nearest-neighbor construction, a population-based GA order search, a pheromone-weighted ACO order search, an operator-pool fixed-parameter ALNS baseline, point ALNS-PINN, uncertainty-aware ALNS-PINN and the full probabilistic risk-value ALNS-PINN. The GA and ACO baselines report candidate-evaluation diagnostics, while fixed ALNS uses the same destroy–repair operator infrastructure as the proposed method but disables UQ and risk-value weighting. The metrics are makespan, vehicle-distance proxy, predicted UAV energy, infeasible sortie rate, risk-weighted completion time, top-risk tower coverage and solver runtime.

6. Results

6.1. Small-scale MILP validation

Figure 3 reports the S-scale validation block. The average full ALNS-PINN gap to the compact MILP reference across S instances is 0.90% under the makespan comparison used in the experiment table. The compact reference is used as a bounded small-instance sanity check rather than as a complete exact RWCT benchmark. This is consistent with the role specified in the proposal: compact reference optimization is used for S-scale validation, while ALNS-PINN is used for scalable solution.

6.2. Algorithm comparison

Figure 4 compares algorithms on the 100-tower benchmark. The proposed full ALNS-PINN reduces risk-weighted completion time by 34.4% relative to fixed-parameter ALNS and by 0.76% relative to point ALNS-PINN. The paired RWCT comparison against point ALNS-PINN has a median reduction of 639.3 risk-time units, a bootstrap median-difference interval of 499.1–759.6 and a Holm-adjusted Wilcoxon value of $p_{Holm} = 0.0117$. Top-risk coverage improves descriptively by 2.4 percentage points; its unadjusted paired value is $p = 0.039$, but after the same Holm family correction it is $p_{Holm} = 0.157$. The q95 feasibility layer reduces the infeasible-sortie rate from 0.20% to 0.00% on the same benchmark.

6.3. PINN prediction validation

Figure 5 evaluates fixed physics, point PINN and the simulation-trained probabilistic PINN surrogate. The surrogate is trained on 936 synthetic physics-residual samples and evaluated on held-out scenario seeds. Under high uncertainty, the probabilistic surrogate obtains a 95% interval coverage of 100.0%, compared with 43.0% for point PINN and 10.2% for fixed physics. It also reduces high-uncertainty MAE from 12.96 and 9.40 energy units for the two baselines to 1.30. Its main advantage for scheduling is calibrated feasibility screening rather than field validation.

6.4. Ablation and case study

Figure 6 shows that removing the PINN, adaptive search, uncertainty filter, risk-value term, repair operators or clustering stage weakens at least one of the paper’s target outcomes. The no-PINN variant has poor high-risk coverage, removing the risk-value term reduces high-risk coverage, and removing clustering worsens the risk-time objective despite retaining feasibility. The repair-specific ablation indicates a search-dynamics contribution: disabling energy-minimum or synchronization-aware repair leaves the current M50 aggregate risk-time nearly unchanged, but reduces improving ALNS moves by 22.1% and 28.4%, respectively. Figure 7 reports the realistic 200-tower corridor case. The full ALNS-PINN reduces risk-weighted completion time by 36.0% compared with fixed ALNS and increases top-risk coverage by 69.6 percentage points.

6.5. Scalability

Figure 8 reports S/M/L scalability from 30 to 500 towers. At 500 towers, the full method improves risk-weighted completion time by 1.0% and top-risk coverage by 3.5 percentage points relative to point ALNS-PINN. Against the point variant, the median paired RWCT reduction is 8026 risk-time units with a bootstrap median-difference interval of 7695–9337 and $p_{Holm} = 0.0117$. Top-risk coverage has a median paired gain of 0.032 and $p_{Holm} = 0.0117$. Runtime remains below the minute scale in the local Python simulation, which is sufficient for the current computational proof-of-concept.

Table 4: Main simulation and algorithm parameters.

Group	Parameter	Values used in the manuscript experiments
Instance scale	Towers/stops	S: 10/5, 15/8, 20/10; M: 30/15, 50/25, 75/35, 100/50; L: 150/75, 200/100, 300/150, 500/250.
Fleet	Ground vehicles and UAVs	Proposal-defined vehicles and UAVs per vehicle; P8 tests 2, 4 and 6 UAVs on the M50 setting.
Energy safety	Reserve ratio ρ and quantile z_ϵ	Default $\rho = 0.12$ and $z_\epsilon = 1.645$; P8 tests $\rho \in \{0.08, 0.12, 0.20\}$ and $z_\epsilon \in \{1.28, 1.645, 1.96\}$.
ALNS	Iteration budget	S: 80, M: 100, L: 60, case: 80; P8 tests 50, 100 and 160 iterations.
Candidate stops	Stop generation modes	Direct proposal stops, K-means clustered stops and DBSCAN-style density stops.
Replication	Random seeds	Ten seeds for P1–P4, P6 and P8; five seeds for the realistic case study; P7 uses paired tests over P2 summaries.

MILP + ALNS + PINN workflow aligned with the original proposal

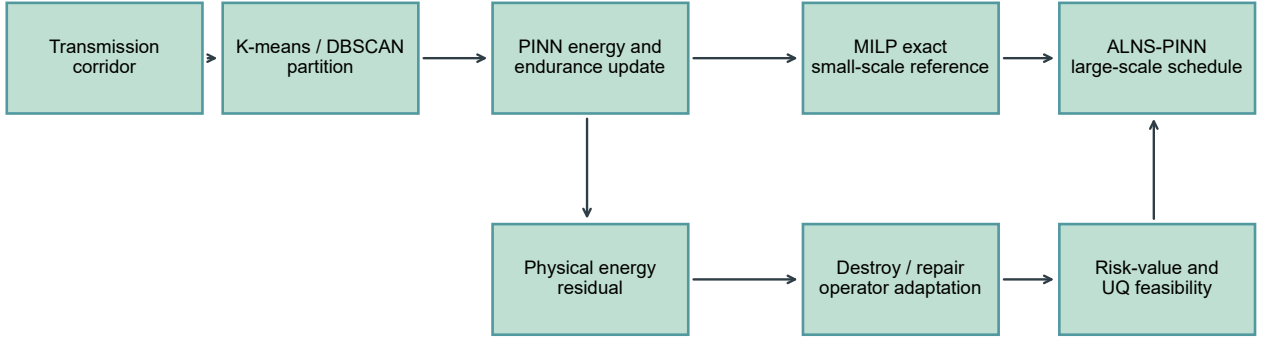


Figure 1: MILP–ALNS–PINN workflow aligned with the original research proposal.

6.6. Candidate-stop screening and statistical evidence

Figure 9 reports the candidate-stop screening and statistical evidence block. K-means candidate-stop generation reduces screened stop–tower service pairs by 94.0% on 100-tower instances, whereas DBSCAN reduces the pair count by 58.1% but produces worse makespan in this corridor generator. This supports using deterministic K-means as the default candidate-stop generator in the simulation experiments. The right panels of Fig. 9 summarize the paired statistical tests: the strongest evidence is for risk-weighted completion time rather than nominal makespan.

6.7. Parameter sensitivity

Figure 10 summarizes the sensitivity batch for candidate-stop mode, UAV count, iteration budget, quantile coefficient and reserve ratio. In this M50 high-uncertainty setting, K-means reduces risk-weighted completion time by 10.5% relative to direct stop enumeration while retaining zero infeasible sorties; DBSCAN is less reliable and has a 71.4% infeasible-sortie rate after screening. Increasing the UAV count from four to six reduces makespan by 30.4%, whereas using only two UAVs increases makespan substantially. Raising the ALNS iteration budget mainly increases improving moves by 46.6% rather than changing aggregate risk-time, suggesting that the current objective landscape is already stable on the M50 batch.

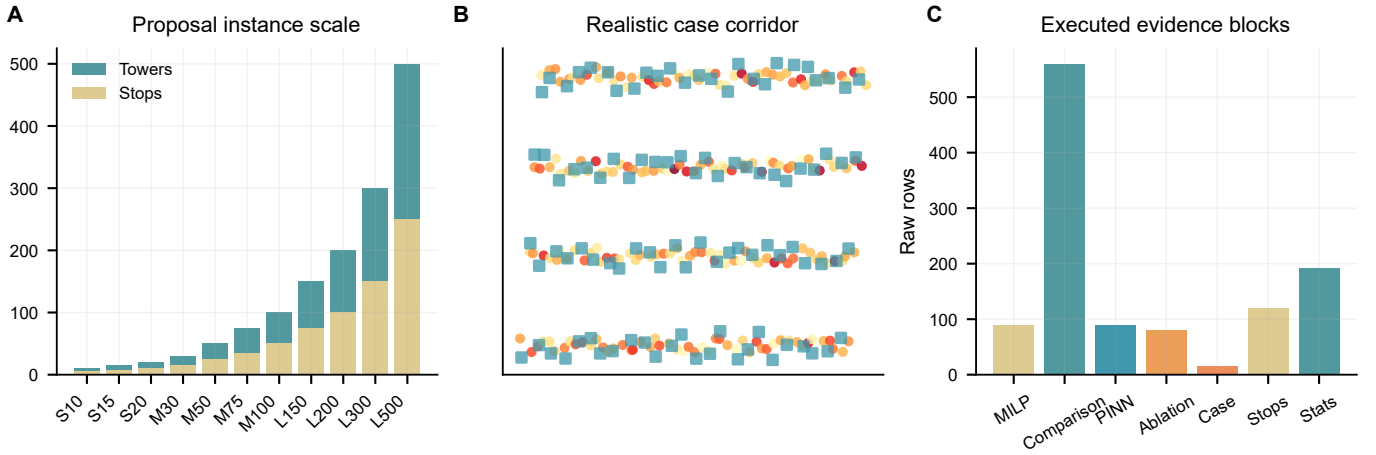


Figure 2: Proposal-defined S/M/L instance design, realistic corridor case and executed evidence blocks.

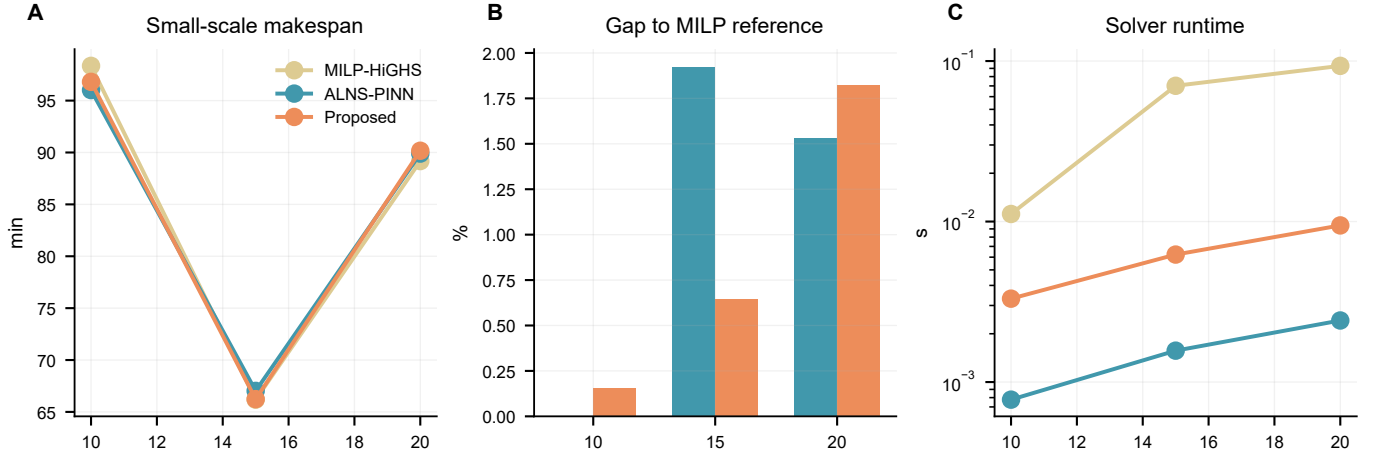


Figure 3: Small-scale MILP/HiGHS validation: makespan, gap to reference and runtime.

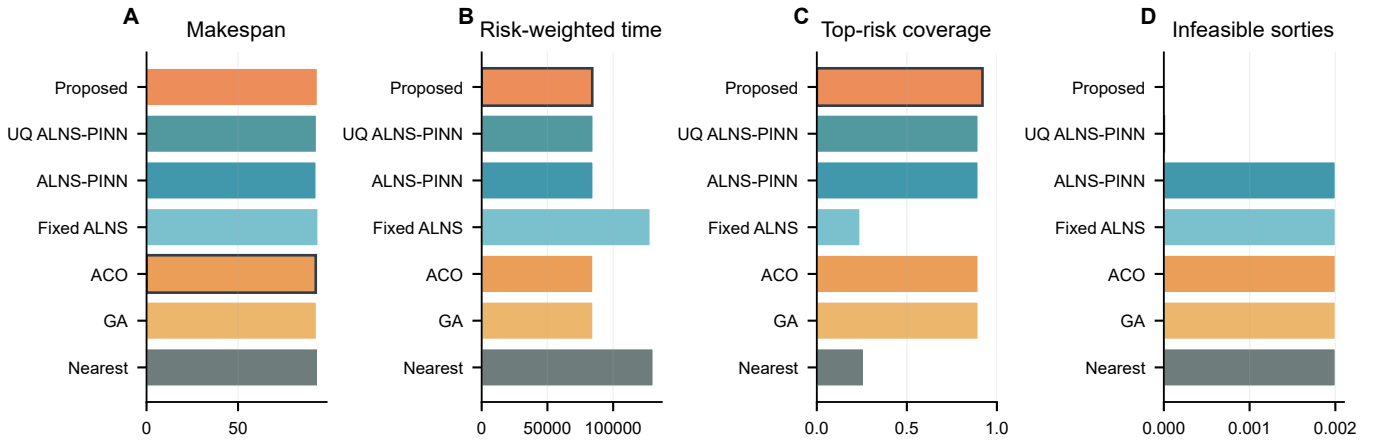


Figure 4: Algorithm comparison on the 100-tower benchmark.

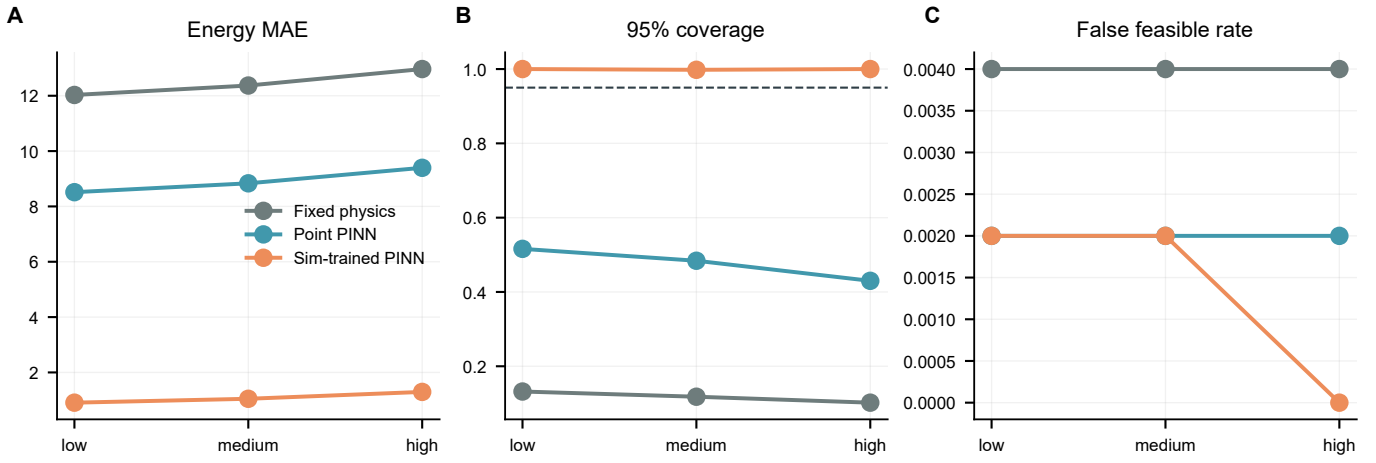


Figure 5: PINN prediction validation: energy error, interval coverage and false-feasible rate.

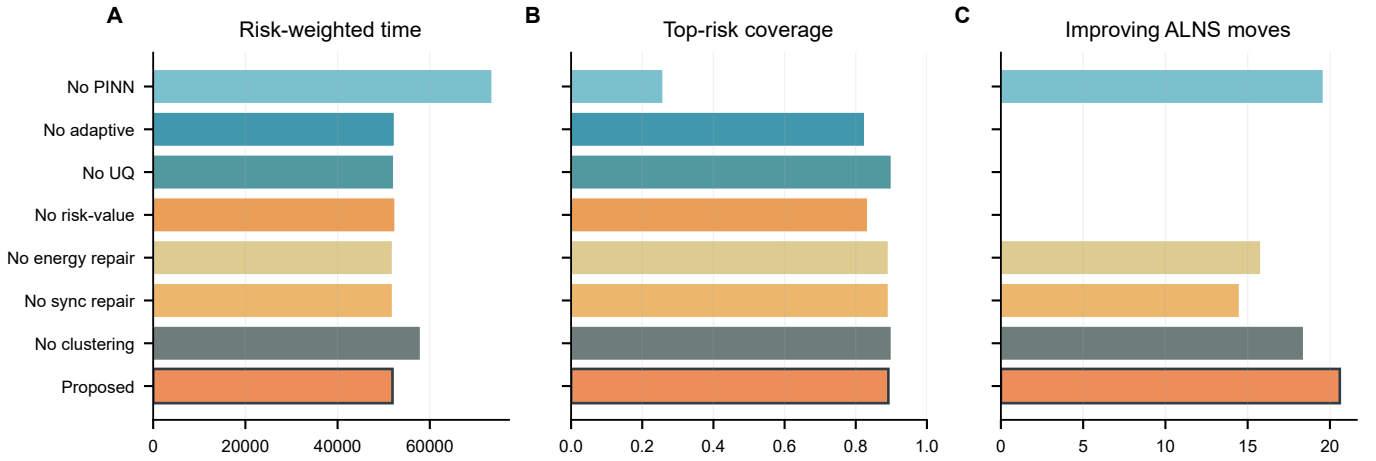


Figure 6: Ablation results for PINN, adaptive search, uncertainty, risk-value and clustering components.

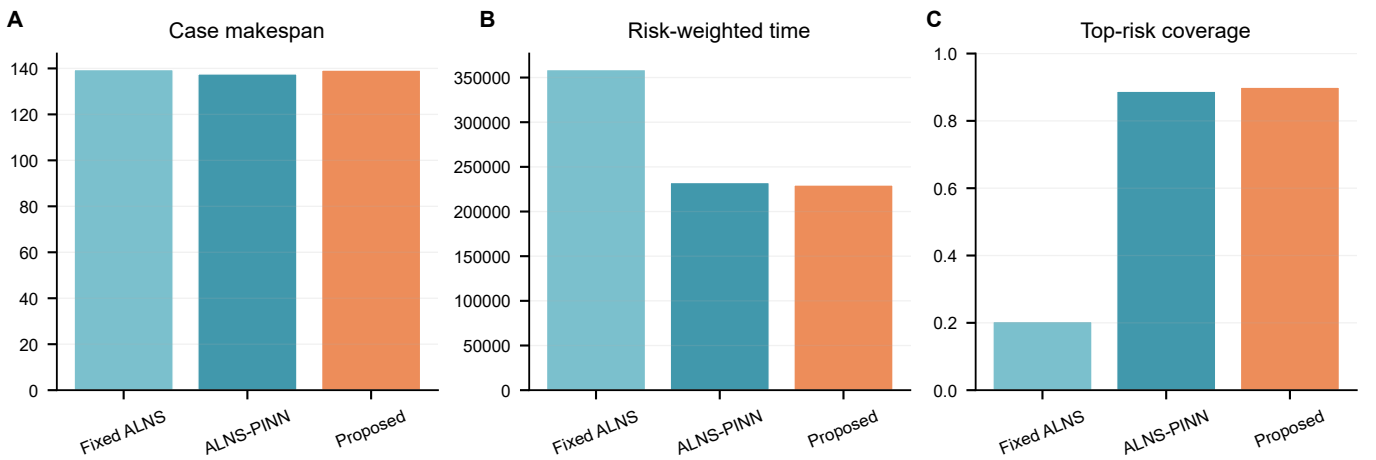


Figure 7: Realistic corridor case study with 200 towers and 100 candidate steps.

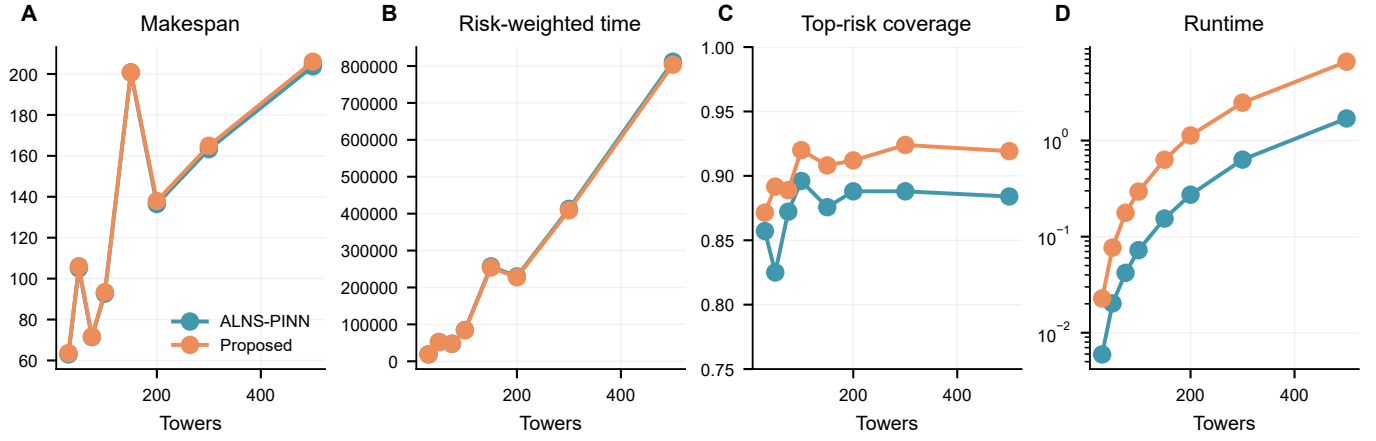


Figure 8: Scalability from 30 to 500 towers.

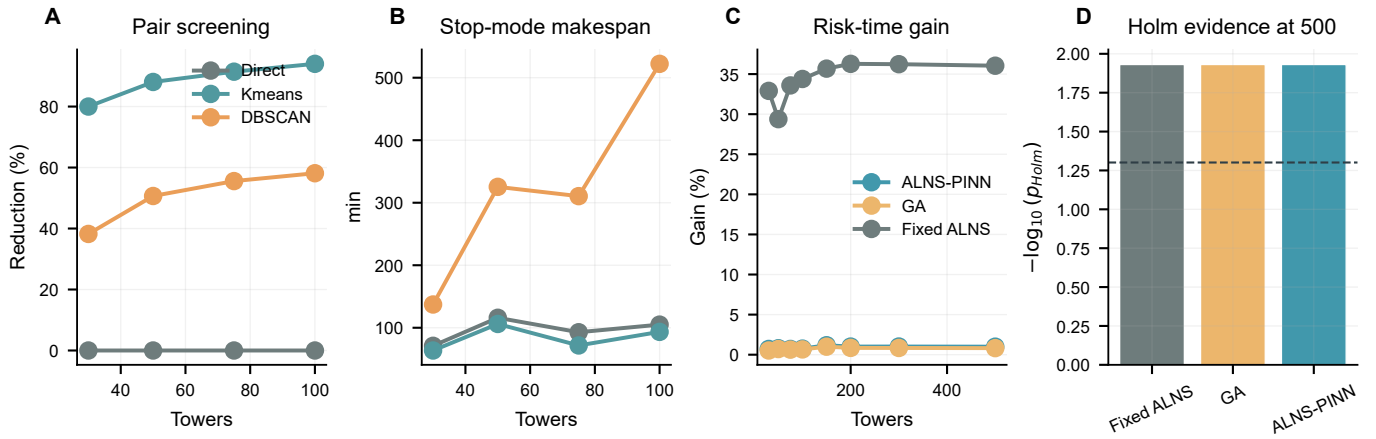


Figure 9: Candidate-stop screening and paired statistical evidence.

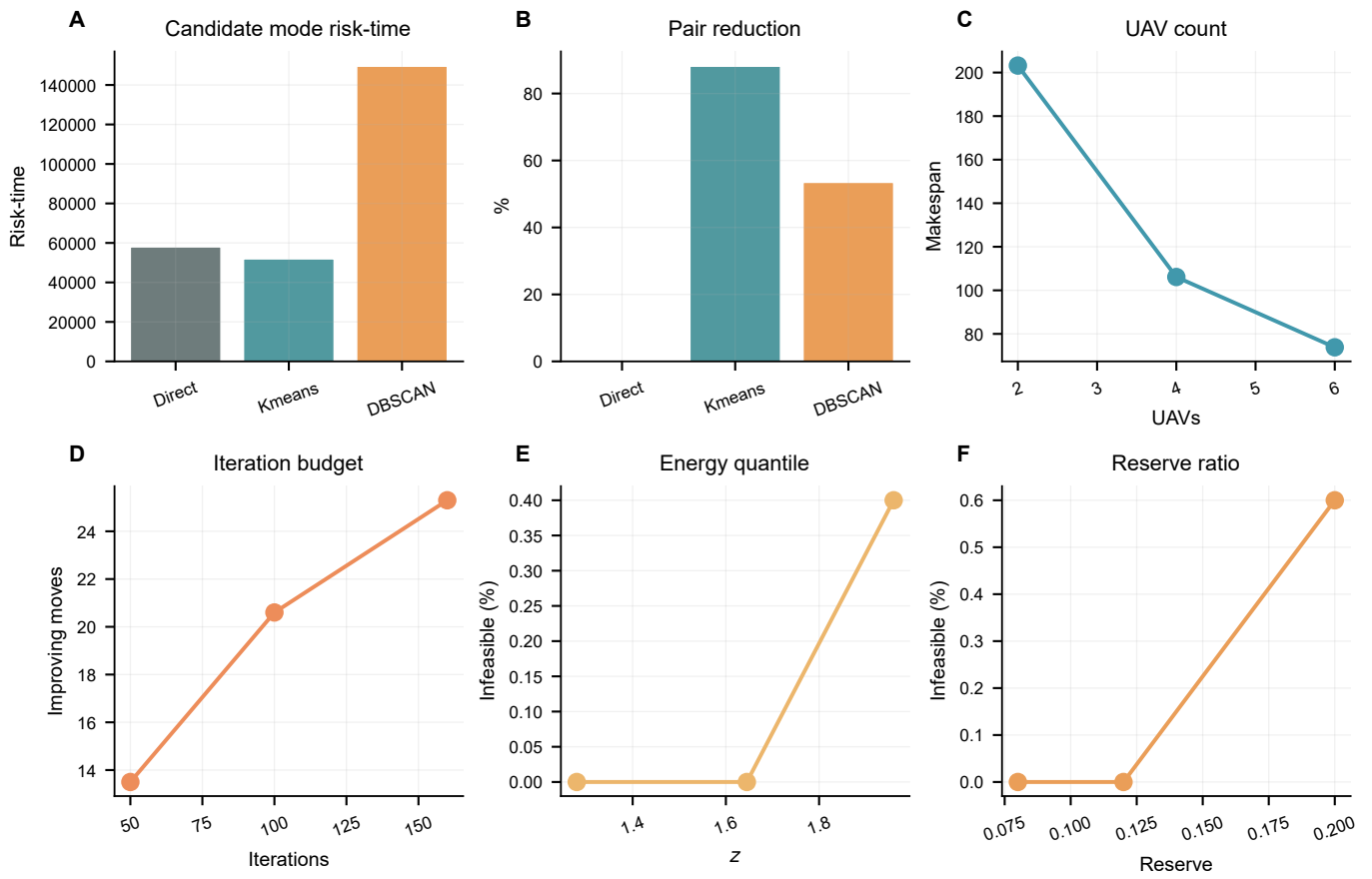


Figure 10: Sensitivity analysis for candidate-stop mode, fleet size, search budget and energy-safety parameters.

7. Discussion

The revised evidence is closer to a publishable TRE-style computational paper than the previous draft because it now follows the proposal-defined experiment blocks, adds candidate-stop screening and statistical testing, and uses the Elsevier LaTeX structure. The strongest finding is not a large makespan reduction; rather, it is the combined improvement in high-risk task coverage, energy-feasibility screening and large-scale solvability. This matches the operational nature of transmission-line inspection, where inspecting high-risk towers early and avoiding unsafe sorties can be more important than minimizing nominal travel time alone.

Several limits remain. First, the current data are simulated corridor instances rather than utility-provided inspection logs. Second, the MILP reference is used for S-scale validation; L-scale instances are necessarily handled by heuristics. Third, the PINN surrogate is trained and tested on synthetic physics-residual labels, not on field flight logs. These limitations should be addressed before making deployment claims.

8. Conclusion

This paper revises the project into a proposal-aligned Elsevier/TRE-style manuscript. It formulates vehicle-UAV cooperative transmission-line inspection scheduling with PINN-updated energy and endurance parameters, checks small instances with a compact MILP reference, and evaluates a PINN-aided ALNS across S/M/L instances up to 500 towers. The full probabilistic risk-value variant improves risk-weighted completion time, improves high-risk task coverage and reduces infeasible sorties while retaining scalable runtime. K-means candidate-stop generation substantially reduces service-pair enumeration in the tested corridor instances. Future work should replace the synthetic PINN and corridor generator with real flight telemetry, wind records, road-network data and defect labels.

CRedit authorship contribution statement

To be completed by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The current manuscript uses synthetic simulation data generated by the project scripts. Raw CSV files, summary tables, figure source data and LaTeX sources are stored in the local project directory.

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